ECE750T: Research Project Proposal

Reasoning Under Observation: Observer Effects on LLM Rationales

Yehia Fahmy | 20786266
University of Waterloo | Fall 2026

Q1: Research Question

How do observer identity and the declared visibility of a language model's verbalized reasoning affect answer accuracy, alignment with a misleading anonymous suggestion, and acknowledgment of that suggestion? I will compare a developer, a domain expert, and an automated AI monitor, each declared to see either the rationale and answer or only the answer. Oversight framing may change the evidence an audit collects, connecting evaluation and foundation-model safety.

Q2: Prior Work and Limitations

Turpin et al. [2023] and Chen et al. [2025] show that hints can influence answers without being acknowledged, but do not test this visibility comparison. Chen et al. [2026] study reasoning controllability, including explicit AI-monitoring awareness, rather than hint acknowledgment across observer roles. Covas and Hidalgo Toledo [2026] study framing in one debate topic; their human–AI observer contrast is suggestive rather than statistically resolved. Li et al. [2026] distinguish evaluation cues, recognition, and response; verbalized recognition is often not associated with behavioural change. These motivate a controlled extension, not a new claim of awareness.

Q3: Novelty of Approach

I extend hint-disclosure and monitoring-awareness studies by crossing observer identity with declared rationale visibility under a fixed misleading suggestion on a small local model. It tests framing effects when hints are present, not causal changes in hint susceptibility under every observer setting.

Q4: Impact and Motivation

If monitoring changes a rationale's acknowledgment of external suggestions, auditors may misjudge its transparency. Benchmark designers and safety evaluators could use the results to assess how oversight framing affects chain-of-thought audit evidence in this limited setting.

Q5: Risks and Assumptions

Hint alignment does not prove causal influence, and rationales do not establish internal computation or intentional concealment. Answer-only observer access is declared, not actual privacy or model belief. Matched templates cannot remove role/authority confounding. Separate pilot items will check solvability, trace truncation, and labeling reliability/time. A comprehension probe checks the access instruction, not belief. My sample review will check automated labels; API confidence is not proof of correctness, and observer mentions may compromise blinding. Small samples, benchmark exposure, and one-model generalization limit conclusions.

Q6: Resources

I will download public BIG-Bench Hard data [Suzgun et al., 2023] and Qwen3-4B Q4_K_M from the official BBH and Qwen repositories. My available compute is an 8 GB RTX 3070 Ti. The target is 120 main items (40 per task): 960 generations, plus separate pilot items. Python and `llama.cpp` will record thinking traces and answers with the same format, thinking mode, and decoding throughout. Inference stays local, with no training. I will arrange an API key and billing for the OpenAI Decisions API for acknowledgment labels; pilot-test memory, throughput, API access, and cost.

Q7: Weekly Plan

At five hours per week, I budget 34 hours plus four in reserve through the provisional December 4 planning date. The first 8–10 hours check inference, scoring, trace access, API label reliability, and review time. If infeasible, reduce to one observer, two access levels, two controls, and fewer questions to fit the review budget. Extra seeds and paraphrases are optional.

Week	Plan
Oct 12–18 (4h)	Refine literature; acquire data/model; build an AI-assisted inference/scoring smoke test.
Oct 19–25 (2h)	Check formats; draft labeling rules and arrange API access; keep midterm week light.
Oct 26–Nov 1 (5h)	Pilot tasks and Decisions labels; check sample-review time and freeze the protocol.
Nov 2–8 (4h)	Run the selected design and API labeling; check complete traces and saved metadata.
Nov 9–15 (5h)	Check sample labels and uncertain cases; start paired analysis.
Nov 16–22 (5h)	Finish figures and intervals; draft the report and stop new experiments.
Nov 23–29 (4h)	Revise the report and prepare/rehearse slides around the micro-survey deadline.
Nov 30–Dec 4 (5h)	Finalize report, slides, and reproducibility files under final instructions.

Q8: Milestones and Success Criteria

Pilot by November 1. Test Decisions API labels for explicit acknowledgment of hints (currently `gpt-6-luna`, public beta). Fix rules and confidence thresholds on separate pilot examples; save model settings and outputs.

Main study by November 15. Test three observers with two access levels, plus no-observer controls with/without hints. In fresh conversations, keep each random wrong hint fixed across hinted conditions. Score answers directly; use the API for acknowledgment. Independently sample and check 40–60 traces across conditions and both predicted labels, hiding API labels during review; also review uncertain cases. Report false positives and missed acknowledgments (precision/recall); expand checks if needed.

Analysis and report draft by November 22. The main measure is the percentage of all questions per condition where the model chooses the wrong hint without acknowledging it. Compare access levels, averaging over roles; role comparisons are exploratory. Report accuracy, hint agreement, and acknowledgment. Compare no-observer controls using the same wrong label (hidden when no hint is given). Estimate uncertainty with 95% bootstrap intervals by resampling matched questions within tasks (one seed schedule). Success means reproducible results, checked scoring, measured label reliability, and clear limitations; small effects may remain unresolved even if no difference is found.

Tools and Assistance

I used Cursor-supported models for literature discovery, planning, and drafting. Cursor and ChatGPT/Claude will generate and debug scripts and assist analysis; the Decisions API will label acknowledgment. I will review and run scripts, verify citations and results, retain sample checks, and disclose tool use.

References

Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, Vlad Mikulik, Samuel R. Bowman, Jan Leike, Jared Kaplan, and Ethan Perez. Reasoning models don't always say what they think, 2025. URL https://arxiv.org/abs/2505.05410.

Yueh-Han Chen, Robert McCarthy, Bruce W. Lee, He He, Ian Kivlichan, Bowen Baker, Micah Carroll, and Tomek Korbak. Reasoning models struggle to control their chains of thought, 2026. URL https://arxiv.org/abs/2603.05706.

Vinicius Covas and Jorge Alberto Hidalgo Toledo. Ai knows when it's being watched: Functional strategic action and contextual register modulation in large language models, 2026. URL https://arxiv.org/abs/2605.15034.

Changling Li, Terry Jingchen Zhang, Jie Zhang, Zhijing Jin, Florian Tramèr, Sahar Abdelnabi, and Maksym Andriushchenko. Decomposing and measuring evaluation awareness, 2026. URL https://arxiv.org/abs/2605.23055.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc Le, Ed Chi, Denny Zhou, and Jason Wei. Challenging BIG-bench tasks and whether chain-of-thought can solve them. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13003–13051. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.findings-acl.824. URL https://aclanthology.org/2023.findings-acl.824/.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://arxiv.org/abs/2305.04388.